Oli Ahammed Sarker

PhD Applicant | Computer Science and Engineering

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LinkedIn | GitHub | Google Scholar | Portfolio

Research Profile

Computer Science and Engineering graduate with three co-authored publications addressing IoT-based precision agriculture, AI applications in cardiovascular health, and blockchain security in banking. My academic research exposure has led me to an interest in how data-driven methods can help investigate real-world problems. Alongside this background, I have professional software engineering experience in backend systems, third-party API integration, databases, and performance optimization, and I aim to pursue doctoral research applying machine learning to agricultural and environmental data.

Research Interests

Machine learning for environmental and agricultural applications; data-driven modeling; AI for scientific and societal problems; IoT and data-intensive systems.

Education

B.Sc. in Computer Science and Engineering

Daffodil International University, Bangladesh CGPA: 3.15/4.00
Graduation: September 2023

Undergraduate Thesis: *Self-Controlled Social Network Approach Using Blockchain Technology*. Designed and built a decentralized social networking application with Solana, Anchor, IPFS and Web3.Storage and a React interface. A manuscript based on this work is available on ResearchGate (not peer-reviewed).

Publications

1. Md Redwan Hussain, Jarin Tias Meraj, and **Oli Ahammed Sarker**. “Precision Farming Through the Use of Internet of Things (IoT) Innovations in Agriculture.” *Journal of Sustainable Agricultural Economics*, 1(1), 18–21, 2024. DOI.
2. Md Shafiqul Islam, Md Mesbah Uddin, Md. Firoz Hossain, Md Redwan Hussain, **Oli Ahammed Sarker**, Toufiqur Rahman Tonmoy, and Rokeya Khatun Shorna. “AI-Powered Early Detection of Cardiovascular Diseases: A Global Health Priority.” *Journal of Advances in Medical Sciences and Artificial Intelligence*, 1(1), 19–25, 2024. DOI.
3. Syed Nazmul Hasan, Mst. Khadijatul Kubra Shinfa, **Oli Ahammed Sarker**, Md Redwan Hussain, and Jarin Tias Meraj. “Blockchain Based Security Solutions for Banking Information Technology.” *Transactions on Banking, Finance, and Leadership Informatics*, 1(1), 25–29, 2024. DOI.

Professional Experience

Software Engineer — Air Valencia Ltd., Dhaka May 2026 – Present

- Work on a production B2C/B2B flight-booking platform (PHP, Laravel, MySQL) integrating Sabre, Amadeus and Verteil NDC supplier APIs.
- Parse, validate and transform large, differently structured supplier responses into one consistent internal format across search, pricing and booking workflows.
- Reduced flight-search response time from about 30–40 seconds to 3–4 seconds using caching, chunked responses and background processing of non-critical tasks.
- Handle API debugging and production support for a live, high-availability system.

Software Engineer — Timestec IT Solutions, Narayanganj Apr 2025 – Apr 2026

- Worked on school-management software used by more than 40 institutions, covering attendance, results, fees and SMS workflows.
- Maintained backend features, REST APIs and database operations for school-management and real-estate applications; supported deployment and user acceptance testing.

Junior Software Engineer — Arena Phone BD Ltd., Dhaka Jul 2024 – Mar 2025

- Worked on a Vehicle Management System for an organization with more than 11,000 employees, handling high-traffic concurrent requests.
- Built SMS-based attendance processing: received messages through an API, sanitized and validated them, and counted attendance only for messages in the required format.
- Worked with the system's GPS-based geofencing module, which detects and counts vehicles entering defined areas.
- Improved application performance and security through query optimization, efficient data processing, and input validation and sanitization.

Technical Skills

Programming: PHP, Python, JavaScript, SQL, Java, C
Machine Learning: supervised and unsupervised learning (regression, classification, clustering), decision trees and ensemble methods, neural networks, data preprocessing, feature engineering, model evaluation and validation
Python Data Stack: NumPy, Pandas, Matplotlib, Seaborn, scikit-learn, PyTorch, Jupyter
Backend and Data Engineering: Laravel, CodeIgniter, REST APIs, JSON/XML parsing, data validation, query optimization, caching, background jobs
Databases: MySQL, MariaDB
Tools: Git, Docker, Postman, Linux, Composer

English Language Proficiency

IELTS Academic: Overall 6.5 (Listening 6.0; Reading 6.0; Speaking 6.5; Writing 6.5).

References

Three academic references are available on request.